

Course Code: CL2005	Course Name: Database Systems Lab
Instructor Name(s): Mr. Sohail Ahmed, Muhammad Nadeem Ghouri	
Student ID:	Section:
Total Time: 90 Minutes	Total Points: 50 points

Instructions:

- Return the question paper mention your **student ID on it**.
- Read each question completely before answering it. **There are 3 questions on 2 pages.**
- Attempt all the given questions. **All questions are carrying different points.**
- **Cheating in any case will lead to F-GRADE as per university rule.**
- In case of any ambiguity, you may make assumption. But your assumption should not contradict any statement in the question paper.

Question # 1: SQL Queries**[10 Points]**

1. Write a query to calculate the total salary cost for each job title and sort the results in descending order of total salary cost.
2. Write a query to find departments with total salary cost above \$70000.
3. Write a query to find out the given output.

	FIRST_NAME	LAST_NAME	JOB_ID	PHONE_NUMBER	DEPARTMENT_NAME	CITY	STATE_PROVINCE
1	Jennifer	Dilly	SH_CLERK	650.505.2876	Shipping	South San Francisco	California
2	Trenna	Rajs	ST_CLERK	650.121.8009	Shipping	South San Francisco	California
3	Jennifer	Whalen	AD_ASST	515.123.4444	Administration	Seattle	Washington
4	Hermann	Baer	PR_REP	515.123.8888	Public Relations	Munich	Bavaria

4. Write a query to display the total number of employees in each department and sort the results by the number of employees in descending order.
5. Write a query to display the first name, email, department name, job id and salary of the most junior employee in each department.
6. Write a query to find out the given output.

	JOB_TITLE	DEPARTMENT_NAME	FULL_NAME	START_DATE
1	Public Accountant	Administration	Jennifer Whalen	01-JUL-02
2	Sales Representative	Sales	Jonathon Taylor	24-MAR-06
3	Programmer	Executive	Lex De Haan	13-JAN-01
4	Accounting Manager	Executive	Neena Kochhar	28-OCT-01

7. Write a query to calculate the average salary for each job title.
8. Write a query that will identify all employees who work in departments located in Germany.
9. Write a query to display the detail information of those departments which starting salary is at least 10000.
10. Write a query to find out the given output.

	FIRST_NAME	LAST_NAME	SALARY	EMAIL	DEPARTMENT_ID	DEPARTMENT_NAME
1	David	Austin	4800	DAUSTIN	60	IT
2	Bruce	Ernst	6000	BERNST	60	IT
3	Pat	Fay	6000	PFAY	20	Marketing
4	Michael	Hartstein	13000	MHARTSTE	20	Marketing
5	Alexander	Hunold	9000	AHUNOLD	60	IT
6	Diana	Lorentz	4200	DLORENTZ	60	IT
7	Valli	Pataballa	4800	VPATABAL	60	IT

Question # 2: DDL**[20 Points]**

You are tasked with designing a database for a movie rental company. The company wants to keep track of information about movies, customers, rentals, and reviews. Create the necessary database tables with the following requirements, create tables and perform the specific Queries.

Movies Table

- a) Each movie should have a unique identifier.
- b) Store the title, release year, and genre of each movie.
- c) Ensure that the title cannot be null.
- d) Ensure that the release year is a 4-digit number (e.g., 2000).
- e) Ensure that the genre is one of the following: 'Action,' 'Drama,' 'Comedy,' 'Horror,' 'Sci-Fi,' or 'Fantasy.'

Customers Table

- a) Each customer should have a unique identifier.
- b) Store the customer's first name, last name, email address, and date of birth.
- c) Ensure that the first name and last name cannot be null.
- d) Ensure that the email address is in a valid format (e.g., "customer@example.com").
- e) Ensure that the date of birth is not in the future.

Rentals Table

- a) Each rental should have a unique identifier.
- b) Store the rental date, return date, and the associated movie and customer.
- c) Ensure that the rental date is not in the future.
- d) Ensure that the return date is after the rental date.
- e) Use foreign keys to link each rental to a specific movie and customer.

Reviews Table

- a) Each review should have a unique identifier.
- b) Store the review text, rating (on a scale of 1 to 5), and the associated movie and customer.
- c) Ensure that the rating is between 1 and 5.
- d) Use foreign keys to link each review to a specific movie and customer.

Queries:**Inserting Records**

- a) Insert a new movie record with the title "Avengers: Endgame," released in 2019, and genre "Action."
- b) Insert a new customer record with the first name "Sarah," last name "Smith," email "sarah@example.com," and date of birth "1985-03-10."

Updating Records

- a) Update the genre of the movie with the title "Avengers: Endgame" to "Sci-Fi."
- b) Update the email address of the customer with the first name "Sarah" to "sarah.newemail@example.com."

Deleting Records

- a) Delete the movie record with the title "Avengers: Endgame."
- b) Delete the customer record with the first name "Sarah."

Altering Tables

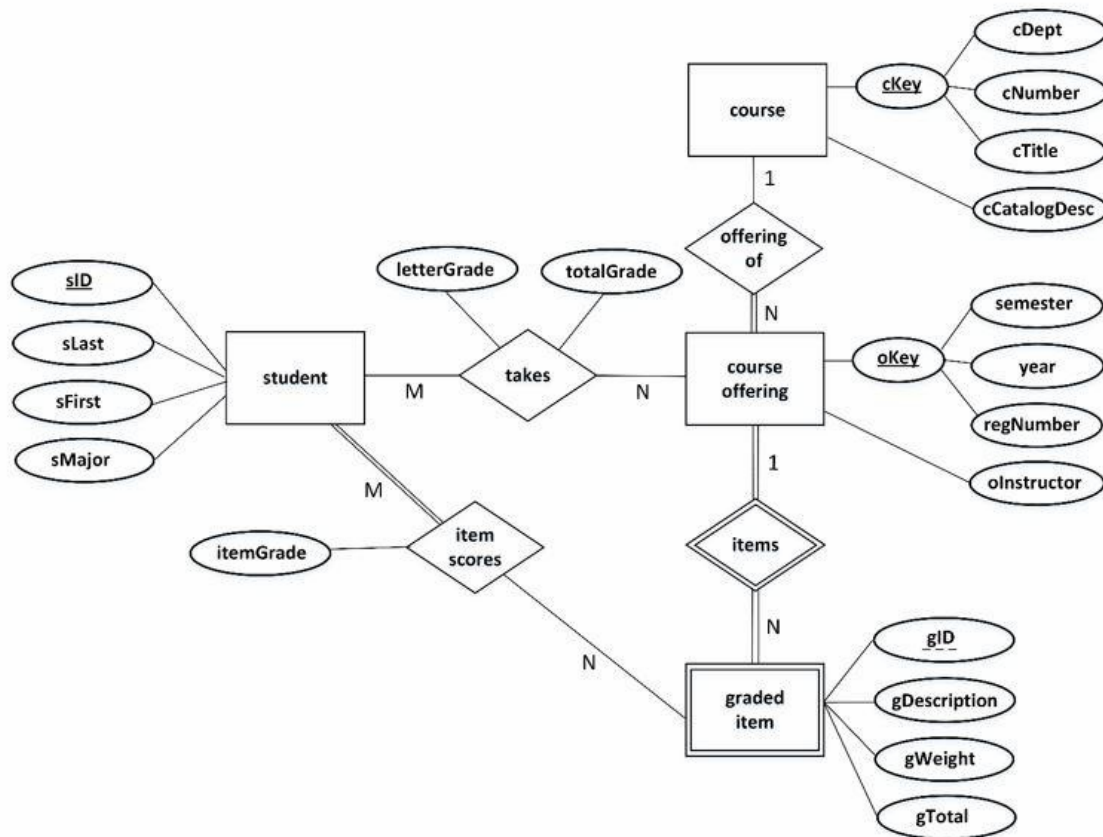
- a) Add a new column named 'director' with the VARCHAR (255) data type to the Movies table.
- b) Add a foreign key constraint to the Rentals table, linking the 'movie_id' column to the 'movie_id' column in the Movies table.

Retrieving Data

- a) Retrieve all movies in the 'Action' genre.
- b) Retrieve all rentals made by a specific customer.
- c) Retrieve movie details along with their reviews, including the movie title, review rating, review text, and customer's first and last name.

Question # 3: Relational Modeling

[20 Points]



- Develop the Logical Model.
- Develop the Relational Model.
- Generate DDL.

“If there is something, you don’t know today. You will surely learn afterwards. Life is not an exam hall.”

BEST OF LUCK ☺